

Build Job Log: Selective Laser Melting of 316L Stainless Steel Flat Washers

Job Information

Job ID: WAS-316L-240520-01

Date: May 20, 2024

Operator: M. Chen (Shift B)

Machine: SLM Solutions 280 HL

Build Platform: Standard 280mm × 280mm

Part Description: ASTM A276 316L stainless steel flat washers

Target Quantity: 33 pieces

Scheduled Duration: 14 hours (including setup and cooldown)

Pre-Build Checklist Completed: 07:45

- Safety systems verified
- Argon supply pressure: 185 bar (within operating range)
- Process water circulation: Normal
- Powder sieving system: Clean and operational
- Build platform: Leveled and cleaned
- Laser calibration: Within specification

Build Preparation and Material Loading

Timestamp: 08:00

Activity: Powder loading and system initialization

Fresh X2CrNiMo1712 stainless steel powder for atomization: 4.11 kg loaded into feedstock hopper.

Process water for water atomization system: 16.8 kg measured and added to cooling circuit.

Timestamp: 08:20

Activity: Chamber purging sequence initiated

Argon chamber filling volume: 700 L standard procedure

Argon consumption rate during build: 54 L per component

Build file loaded: "Washer_316L_33pc.sli"

Layer thickness: 30 µm

Laser scan strategy: Stripes with 67° rotation

Build Process Execution

Timestamp: 08:45

Activity: Build sequence started

Chamber oxygen level: 78 ppm (within specification <100 ppm)
Build platform preheat temperature: 80°C

Process Parameters: - Processing time per working cycle: 13.38 hours - Laser nominal power when active: 5.5 kW - Machine nominal power when laser off: 3.5 kW - Number of components per SLM working cycle: 33 pieces

Timestamp: 09:15
Activity: First layers completed
Laser power stability: Within ±2%
Powder recoating: Normal operation
Chamber temperature: 32°C

Timestamp: 12:30
Activity: Mid-build inspection
Layer 450 of 1,120 completed
Powder bed quality: Uniform, no irregularities
Laser focus: Maintained within tolerance

Timestamp: 16:00
Activity: Process monitoring
Chamber oxygen level: 82 ppm
Cooling water temperature: 24°C
Powder feed rate: Consistent

Timestamp: 20:15
Activity: System status check
Argon supply pressure: 162 bar
Process water level: Normal
Vibration sensors: All within green range

Timestamp: 22:05
Activity: Build completion
Final layer deposited
Total active processing time: 13.38 hours
Laser operating hours logged: Machine total now 1,847.3 hours

Material Consumption and Usage Data

Raw Materials Input:

Material Type	Quantity	Purpose
X2CrNiMo1712 stainless steel powder	4.11 kg	Feedstock for water atomization
Process water	16.8 kg	Cooling media for atomization

Gas Consumption Parameters: - Argon volume per component: 54 L - Number of components in build: 33 pieces - Chamber filling volume: 700 L - Total argon volume calculated from above parameters

Energy Usage Parameters:

Water atomization process: - Energy consumption rate for melting: 2.23 MJ per kg of material processed - Material processed: 4.11 kg

SLM process energy parameters: - Processing duration: 13.38 hours - Machine power with laser active: 5.5 kW - Machine power with laser inactive: 3.5 kW

Reference: Previous similar job (April 15, 2024) used 13.5 hours processing time with comparable power settings.

Product Output and Quality

Timestamp: 22:30
Activity: Build platform removal and initial inspection

Finished Parts Data:

Parameter	Value
Total parts produced	33 units
Total mass of finished washers	0.61 kg
Visual inspection pass rate	100%
Dimensional verification	Within drawing tolerance

First Article Measurement: - Outer diameter: 19.05 mm $\pm$ 0.05 mm - Inner diameter: 9.53 mm $\pm$ 0.05 mm - Thickness: 1.52 mm $\pm$ 0.03 mm - Surface roughness: Ra 12-15 μ m (as-built condition)

Material Recovery and Waste Management

Timestamp: 23:00
Activity: Powder recovery and sieving

Recovered Materials:

Material Type	Quantity	Disposition
316L powder reused in SLM process	2.94 kg	Returned to process hopper
316L powder returned for remelting	0.15 kg	Sent to water atomization
Recovered process water	16.4 kg	Returned to cooling system

Waste Streams:

Waste Type	Quantity	Disposition
Solid waste from water atomization	0.41 kg	Landfill disposal
Non-recyclable 316L powder	0.01 kg	Landfill disposal

Powder Recycling Efficiency: 94.2% of unused powder recovered for reuse

Note: Industry benchmark for powder reuse typically 90-95% for 316L applications.

Equipment Performance and Maintenance Notes

Machine Performance Summary: - Laser uptime: 100% (no interruptions) - Powder delivery system: No blockages - Recuperation system: Normal operation - Cooling system: Stable temperature control

Maintenance Performed: - Filter replacement: Not required this cycle - Lens cleaning: Scheduled after next build - Rail lubrication: Completed during setup

Alarms and Events: - 14:22: Minor fluctuation in powder feed sensor (auto-corrected) - 19:45: Temporary chamber temperature rise to 35°C (returned to normal within 2 minutes)

Post-Build Procedures

Timestamp: 23:45

Activity: System shutdown and cleanup

Build platform moved to depowdering station

Chamber cleaned and prepared for next job

Argon supply valve closed

Process water system drained and flushed

Next Maintenance Due: - Laser source: 2,000 hours (152.7 hours remaining) - Filter replacement: Next build - Full calibration: 2,500 hours

Operator Comments and Sign-off

Operator Notes: “Build completed without significant issues. Powder consumption slightly lower than estimated. All parts visually acceptable. Recommend continuing current parameter set for future washer production.”

Quality Control: Parts sent to post-processing for stress relief and inspection.

Build Status: COMPLETED SUCCESSFULLY

Signed: M. Chen, Operator

Date: May 21, 2024 00:15

Verified: A. Rodriguez, Process Engineer

Date: May 21, 2024 08:30

End of Build Job Log WAS-316L-240520-01